Roll No……..

MID SEMESTER EXAMINATION MARCH – 2025

Name of the Program: M.Tech (ME)

Semester: II

Name of the Course: Modeling and Simulation

Paper Code: DEME 44

Time: 90 Minutes

Maximum Marks: 50

Note:

(i) Answer ***all the questions*** by choosing ***any one of the sub questions***.
(ii) Each question carries 10 marks.

Q1	(10 marks)	
(a)	Describe the significance of complex numbers and their applications in engineering analysis?	**CO1**
	OR	
(b)	Find the Laplace transform of $f(t) = t^4 + 7e^{-3t} + \cos 5t$	**CO1**
Q2	(10 marks)	
(a)	Explain the steps involved in solving a linear time-invariant (LTI) system using Laplace transforms?	**CO1**
	OR	
(b)	Find the magnitude and phase of $z = -3 + 4j$. Also, convert $z = -2 - j$ into polar form.	**CO1**
Q3	(10 marks)	
(a)	Explain the role of mass, spring, and damper in mechanical systems? Derive the governing equation for a mass-spring-damper system in translational motion?	**CO2**
	OR	
(b)	What is a block diagram? Explain its importance in control systems and derive the block diagram representation of a second-order mass-spring-damper system?	**CO2**
Q4	(10 marks)	

(a)	Derive the mathematical model of a rotational mass-spring-damper system and explain its significance in dynamic system modeling?	**CO2**
	OR	
(b)	A 5 kg object is subjected to a force of 50 N. If the system has a damping coefficient of 2 Ns/m, find the acceleration of the mass at the instant when its velocity is 10 m/s?	**CO2**
Q5	(10 marks)	
(a)	Explain the principle of energy conservation in dynamic systems? How does energy transfer occur between kinetic energy, potential energy, and damping energy in a mass-spring-damper system?	**CO2**
	OR	
(b)	Define work, energy, and power in mechanical systems? A force of 20 N moves a mass 4 m along a frictionless surface. Calculate the work done by the force?	**CO2**